WATER RESOURCES, WELLS & PUMPS · NOTE WL03

Tube-Well Drilling, Construction, Development, Testing and Sanitary Protection

Course topics: IDE6.2 L13, IDE6.2 L16, IDE-510 T6

Choosing a drilling method

Choose the equipment by the size and depth of the well and the formation to be drilled; it should meet the job, minimise accidents and hazards, and drill economically. Methods: cable-tool (percussion), rotary (straight, air and reverse), hammer drills, core drills and rock-blast units. Every job has three stages: drilling; installing casing and screen; and developing the well.

1. Percussion (cable-tool) drilling

- **Rigs:** light, medium or heavy duty; hand boring sets (tripod, bailers, cutter shoes, anchor bolt set, loading clamp, chain wrenches, pulley blocks, wire rope and small tools) come in 10, 15, 20 and 25 cm sizes and reach about 130 m. Power rigs have a mast and 2–3 hoist lines, a walking beam and an engine, usually truck-mounted.
- **Capacity:** light rigs 15–45 cm diameter to 100–130 m; medium rigs 20–45 cm to 30–300 m; heavy rigs for deep holes, boulders and difficult formations.
- **Principle:** a bit hung on a cable is raised and dropped; its oscillation cuts and loosens the material, which water softens and holds in suspension for the **bailer** (a pipe with a bottom valve and a top ring) to remove.
- **Tool string (top to bottom):** rope socket, jars, drill stem (adds weight and length for straighter cutting) and bit — 1–3 m long, up to 1400 kg, shaped for the formation.
- Use temporary casing to hold the walls; replace it later with the proper casing.
- **Suits** 10–60 cm wells in consolidated rock; works worst in unconsolidated sand and gravel and quicksand; slower than rotary drilling.

2. Direct rotary drilling

- The cutting bit at the bottom of a drill column turns with the column, rotated from the surface. It is the most common tube-well method and the fastest in unconsolidated strata, drilling deep wells up to 45 cm in diameter.
- **Two units:** drilling equipment (derrick or mast, rotating table, hoist, engine) and mud-pumping equipment.
- **Drilling mud** — clay mixed with water in a mud pit, pumped down by a reciprocating pump — lines the walls against cave-ins and carries cuttings up. Keep it a thick, creamy suspension.
- **Bit:** usually three conical toothed wheels, each turning on its own axis, on a heavy drill rod screwed to the **kelly** (square rod) that the rotating table turns while letting it slide down.
- **Pilot hole** (10–15 cm): drill it first to study the formations, find the water-bearing layers and prepare the well log — so you can fix the final depth and strainer positions cheaply before drilling full size.

3. Reverse-circulation rotary drilling

- As direct rotary, but the fluid — clear water — flows the other way: down the annulus and up through the drill pipe with the cuttings, back to a tank where they settle.
- The hole is not cased; hydrostatic pressure of the water and colloidal material on the walls prevent caving. Keep the water pit at least 3 times the hole's displacement volume and always full.
- Needs 2.5 to 7.5 L/s of water — about 5 times as much as direct rotary.
- Drills large holes (≥ 35 cm) fast in soft, unconsolidated formations, 100–180 m deep; used only for gravel-packed tube wells, which develop easily because clear water is used.

4. Rotary air-percussion drilling

- Combines bit rotation, static load and percussion.
- **Down-the-hole hammer:** an air-driven piston hammer just above the bit strikes 600–1200 blows a minute while the bit turns at 15–50 rpm; compressed air lifts the cuttings. Effective in consolidated rock — drills granite 2–3 times faster than air rotary and far faster than percussion.
- **Above-ground (overburden) method:** rotation and percussion applied from the surface, hammer stationary — for slim holes of moderate depth.
- A direct rotary rig can do it with a modified rotary speed, an extra compressor and larger drill pipe: with 14–17 m^3/min at 7 kg/cm^2 it drills 15–20 cm holes to 75 m in consolidated rock. Useful in dry formations and water-scarce arid regions, with high rates in semi-consolidated formations and hard rock.

Construction steps

1. **Well log:** record the materials met, in sequence and by depth, from samples collected continuously and laid out in order. Select the strainer and plain (blank) pipes from it — it is the basis of casing design and screen placement.
2. **Alignment:** keep the well as vertical as possible so pumps go in and out easily, do not stick and do not bend the shafting. Drill the first 6–12 m with great care — it guides the rest; careless drilling lets the tools wander and slant the lower hole.
3. **Casing:** metal tubing in 3 to 4.5 m lengths joined by sockets holds the walls; perforate it opposite permeable aquifers (use a screen where the aquifer has much fine sand) and use blank pipe through non-water-bearing sections.
4. **Gravel packing** (where the aquifer has much fine sand): an envelope 8–15 cm thick of rounded pea gravel from river beds, slightly coarser than the strainer openings (avoid angular or platy gravel). It enlarges the effective diameter, filters fines, guards against cave-ins and fills cavities. In rotary-drilled wells, thin the mud lining with clean water and wash out the excess first — then place the gravel quickly, as thinned mud cannot hold the walls long. Top up as the pack settles.
5. **Development:** remove silt, sand and very fine gravel round the screen to raise permeability and yield — without overdoing it, which can collapse the overburden or crush the well (if sand keeps coming, fit a special screen).
6. **Recuperation (yield) test:** measure the static level; pump at the maximum rate until the level stabilises without excessive drawdown (preferably within 6 m of static); measure the pumping level; the difference is the drawdown. Use the results to choose the pump. Measure levels with a chalked steel tape, an airline and pressure gauge or an electrical depth gauge.

Development method	How it works
Compressed air	Run an air pipe inside a discharge pipe to near the bottom of the screen; build up 7–11 kg/cm² with the air pipe closed, then release it suddenly with a quick-opening valve. The surge alternately raises and drops the pressure, forcing water up. Repeat until clean water flows continuously
Surging	One of the most effective methods: move a plunger up and down in the casing — the upstroke draws water in, the downstroke forces it back out, loosening fines that are then removed. Start slowly, speed up gradually and continue until no sand or mud enters

Locating a well

- Study nearby wells — depth, quality and quantity of water, pumping head (for tube wells, ask the State tube-well organisation).
- Keep it in sanitary surroundings, away from seepage from barns, poultry houses and toilets; above flood level, or capped before floods; as near the point of use as possible. Wells at foothills generally yield more.

Sanitary protection (for drinking water)

1. Keep wells 15–30 m from animal houses, privies, septic tanks and other sources, never downhill of them; follow local health rules; take special care in rock aquifers such as limestone, where pollutants travel farther.
2. Seal the annulus outside the casing with cement grout; fit a watertight cover with only a manhole and pump hole; build a raised concrete apron sloping away from the well; drain waste water away.
3. In bucket-operated wells, ropes and buckets are the main contamination route — give a partial cover and one common rope and bucket instead of separate ones. Better still, an automatic tilting rod empties the bucket into a closed trough while the rope winds on a pulley, so no one touches the water.
4. Disinfect the water whenever contamination is suspected.

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The fastest drilling method in unconsolidated strata is

(a) cable-tool percussion
(b) direct rotary
(c) hand boring
(d) core drilling

Q2. In reverse-circulation drilling, water and cuttings return through the

(a) annular space
(b) drill pipe
(c) casing
(d) bailer

Q3. A down-the-hole hammer delivers about

(a) 60–120 blows/min
(b) 600–1200 blows/min
(c) 6000 blows/min
(d) 15–50 blows/min

Q4. Recommended thickness of a gravel pack is

(a) 1–2 cm (b) 8–15 cm

(c) 30–45 cm (d) 1 m

Q5. Wells for drinking water should be at least ___ from septic tanks and animal houses

(a) 3–5 m (b) 15–30 m

(c) 100 m (d) 300 m

Q6. The first ___ of drilling must be truly vertical because it guides the rest of the well

(a) 1–2 m (b) 6–12 m

(c) 30–40 m (d) 100 m

Answers

Q1 — (b) direct rotary

Q2 — (b) drill pipe

Q3 — (b) 600–1200 blows/min

Q4 — (b) 8–15 cm

Q5 — (b) 15–30 m

Q6 — (b) 6–12 m